

W1 PRACTICE

PART 2 – EXPLORATION

Learning objectives

- Apply type **inference** for variable declarations.
- Handle **nullable** and **non-nullable** variables.
- Differentiate between **final** and **const**.
- Manipulate **strings, lists, and maps**.
- Use **loops** and **conditions** to control flow.
- Define and call functions with positional and **named arguments**, understand **arrow syntax**



LO NO AI

Solve problems entirely on your own.

How to run Dart code?

You can write you code on VS Code, or using this online editor

- [Install Dart and Flutter SDK](#)
- [Online Dart Compiler](#)

Resources for this research

To help you complete this handout, use the following resources:

- | | | |
|-------------------------------|----------------------------------|------------------------------|
| ✓ Variables | ✓ Built-in types | ✓ Conditions |
| ✓ Null Safety | ✓ Lists | ✓ Functions |
| | ✓ Loops | |



EX 4 - Strings, Lists and Maps

CODE : Complete the bellow code to illustrate the concepts:

Strings:

```
// Declare two strings: firstName and lastName and an integer:age
String firstName="Sok";
String lastName="heng";
Int age=12;

// Concatenate the 2 strings and the age
String fullName=firstName+lastName;

// Print result
Print("My name is $fullName, I am $age");
```

Lists:

```
// Create a list of integers
List<int> numbers=[];

// Add a number to the list
Numbers.add(1);

// Remove a number from the list
Numbers.remove(1);

// Insert a number at a specific index in the list
Numbers.insert(0,5);

// Iterate over the list and print each number
Print(number[0]);
```

Maps:

```
// Create a map with String keys and integer values
Var people={"heng":"male", "mesa":"male"};

// Add a new key-value pair to the map
People["bob"]="male";

// Remove a key-value pair from the map
People.remove("bob");

// Iterate over the map and print each key-value pair
Print(people["heng"]);
```

EXPLAIN : When should I use a Map instead of a List?

You use map instead of list when you want to do a key-value pair for each element instead of a single element

EXPLAIN : When should I use a Set instead of a List?

You set when you want each of your element is unique

EX 5 - Loops and Conditions

CODE : Complete the bellow code to illustrate the concepts:

```
// Use a for-loop to print numbers from 1 to 5
For (int i=1;i<=5;i++){
    Print(i);
}

// Use a while-loop to print numbers while a condition is true
Int i=1;
While(i<=5){
    Print(i);
    I++;
}

// Use an if-else statement to check if a number is even or odd
Int num=2;
If(num%2==0){
    Print("even");
}else{
    Print(odd);
}
```

EX 6 - Functions

EXPLAIN : Compare positional and named function arguments

Position function is a function that argument and parameter is in order, named function is a function that position of the argument is not matter.

EXPLAIN : Explain when and how to use arrow syntax for functions?

CODE : Complete the bellow code to illustrate the concepts:

Defining and Invoking a Function:

```
// Define a function that takes two integers and returns their sum

// Call the function and print the result
```

Positional vs Named Arguments:

```
// Define a function that uses positional arguments
Void myfunc(String name,int age){
    Print("Hello my name is $name and I am $age");
}
Void main(){
    myfunc("heng",12);
}

// Define another function that uses named arguments with the required
keyword (ex: getArea with rectangle arguments)
void myfunc({required String name, required int age}) {
    print("Hello my name is $name and I am $age");
}

void main() {
    myfunc(name:"heng",age:12);
}

// Call both functions with appropriate arguments
```

EXPLAIN : Can positional argument be omitted? Show an example

Yes it can

```
Void myfunc([String name],int age){
    Print("Hello my name is $name and I am $age");
}
Void main(){
    myfunc("heng",12);
}
```

```
}
```

EXPLAIN : Can named argument be omitted? Show an example

Yes it can

```
void myfunc({required String name, required int age}) {  
    print("Hello my name is $name and I am $age");  
}  
  
void main() {  
    myfunc(name:"heng",age:12);  
}
```

CODE : Complete the bellow code to illustrate the concepts:

Arrow Syntax:

```
// Define a function using arrow syntax that squares a number  
  
// Call the arrow function and print the result
```